

Multi Tier Computing

Middleware

MIDDLEWARE:

Hardware and Software requirements

- Client/Server software requirements
- 90 percent of information technology (IT) departments' budgets are spent **maintaining and enhancing existing systems**
- **Business Process Reengineering (BPR)** techniques help organizations achieve competitive advantage through substantive improvements in quality, customer service and costs
- BPR must be aligned with technology strategy to be effective.
- Organizations must use technology to enable the business change defined by the BPR

Factors Driving Demand for Application Software Development

- Strategic planning, development, and follow-on support for applications software is a vital,Expensive
- BPR and total quality management (TQM) programs demand software development and enhancements.
- A competitive market insists that companies demonstrate their value to a skeptical buyer through increasing the value of product and services.

Rising Technology Staff Costs

- Technology personnel costs are rising much faster than inflation
- Many organizations find that technology professionals, in whom much organization specific application and technology knowledge has been invested, change jobs every three to five years
- Must develop a continuous learning program

Pressure to Build Competitive Advantage by Delivering Systems Faster

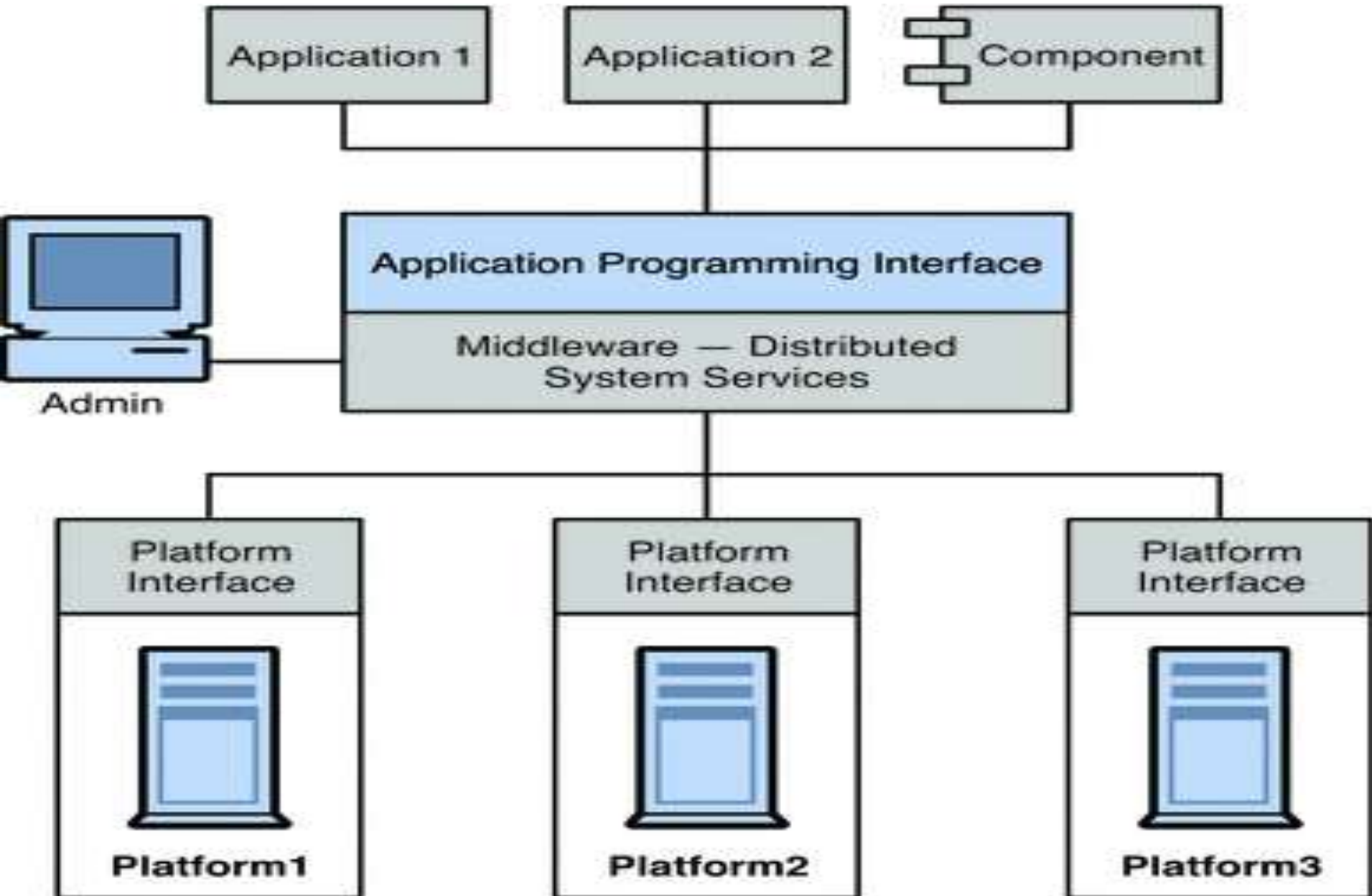
- To build competitive advantage
- The application backlog is horrific
- 80 to 90 percent of the traditional host-based MIS shop's staff time is devoted to maintaining

Need to Improve Technology Professionals' Productivity

- New breed of software tools
- Companies need new, innovative mission-critical systems
 - to be built quickly, with a
 - highly productive, committed professional staff partnered with end-users during the requirements, design, and construction phases.

- **Need for Platform Migration and Reengineering of Existing Systems**
- **Need for a Common Interface Across Platforms**
- **Increase in Applications Development by Users**

Middleware



Middleware

- Easiest way to integrate heterogeneous components is not to recreate them as homogeneous elements but to provide a layer that allows them to communicate despite their differences
- This layer, called middleware, allows software components (applications, enterprise java beans, servlets, and other components) that have been developed independently and that run on different networked platforms to interact with one another

Categories- Middleware

Middleware can be grouped into the following categories:

- 1) **Remote Procedure Call** or RPC-based middleware
- 2) **Object Request Broker** or ORB-based middleware
- 3) **Message Oriented Middleware** or MOM-based middleware

Remote Procedure Call or RPC-based middleware

which allows procedures in one application to call procedures in remote applications as if they were local calls. The middleware implements a linking mechanism that locates remote procedures and makes these transparently available to a caller. Traditionally, this type of middleware handled procedure-based programs; it now also includes object-based components.



Object Request Broker or ORB-based middleware

- Object Request Broker or ORB-based middleware, which enables an application's objects to be distributed and shared across heterogeneous networks.

Message Oriented Middleware or MOM-based middleware

- **Message Oriented Middleware or MOM-based middleware, which allows distributed applications to communicate and exchange data by sending and receiving messages.**

- All these models make it possible for one software component to affect the behavior of another component over a network.
- They are different in that **RPC- and ORB-based** middleware create systems of **tightly-coupled components**, whereas **MOM-based systems** allow for a looser coupling of components.
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- In an **RPC- or ORB-based** system, when one procedure calls another, it must wait for the called procedure to return before it can do anything else.

- In these synchronous messaging models, the middleware functions partly as a super-linker, locating the called procedure on a network and using network services to pass function or method parameters to the procedure and then to return results.
- MOM-based systems allows communication to happen through the asynchronous exchange of messages



THANK YOU